

1 部分正确性注释程序

定义 1.1 (部分正确性注释程序)

对断言 P, Q , 程序 S 和某字符串 S^* , 递归定义

$$\{P\} S^* \{Q\} \vdash_{\text{PO}} \{P\} S \{Q\},$$

称 $\{P\} S^* \{Q\}$ 是 $\{P\} S \{Q\}$ 的**部分正确性注释程序**, $\{P\}, \{Q\}$ 分别为前后断言 (pre, post), S 为骨架程序 (sk):

1. 当 $P \rightarrow Q$ 时, $\{P\} \text{skip} \{Q\} \vdash_{\text{PO}} \{P\} \text{skip} \{Q\}$,
2. 当 $P \rightarrow Q[u := t]$ 时, $\{P\} u := t \{Q\} \vdash_{\text{PO}} \{P\} u := t \{Q\}$,
3. 如果 $\{P\} S_1^* \{R\} \vdash_{\text{PO}} \{P\} S_1 \{R\}$ 且 $\{R\} S_2^* \{Q\} \vdash_{\text{PO}} \{R\} S_2 \{Q\}$, 则

$$\{P\} S_1^* \{R\} S_2^* \{Q\} \vdash_{\text{PO}} \{P\} S_1; S_2 \{Q\},$$
4. 如果 $\{P \wedge B\} S_1^* \{Q\} \vdash_{\text{PO}} \{P \wedge B\} S_1 \{Q\}$ 且 $\{P \wedge \neg B\} S_2^* \{Q\} \vdash_{\text{PO}} \{P \wedge \neg B\} S_2 \{Q\}$, 则

$$\begin{aligned} & \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\} \\ & \vdash_{\text{PO}} \{P\} \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } \{Q\}, \end{aligned}$$
5. 如果 $P \rightarrow I$, $\{I \wedge B\} S^* \{I\} \vdash_{\text{PO}} \{I \wedge B\} S \{I\}$ 且 $I \wedge \neg B \rightarrow Q$, 则

$$\begin{aligned} & \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\} \\ & \vdash_{\text{PO}} \{P\} \text{while } B \text{ do } S \text{ od } \{Q\}. \end{aligned}$$

定理 1.1 (正规形定理)

对任意的断言 P, Q 和程序 S , $\vdash_{\text{PW}} \{P\} S \{Q\}$ 当且仅当存在 $\{P\} S^* \{Q\} \vdash_{\text{PO}} \{P\} S \{Q\}$.

定义 1.2 (部分正确性注释程序上的操作)

把 $\{P\} \downarrow \{P\}$ 也视为自身的注释程序, 递归定义注释程序带状态的一步转移关系:

1. $\langle \{P\} \text{skip} \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} \downarrow \{Q\}, \sigma \rangle$,
2. $\langle \{P\} u := t \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} \downarrow \{Q\}, \sigma[u := \sigma(t)] \rangle$,
3. 任给注释程序 $\{P\} S_1^* \{R\}$ 和 $\{R\} S_2^* \{Q\}$, 如果 $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{P'\} S_1'^* \{R\}, \sigma' \rangle$, 那么

$$\langle \{P\} S_1^* \{R\} S_2^* \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P'\} S_1'^* \{R\} S_2^* \{Q\}, \sigma' \rangle,$$
4. 任给注释程序 $\{P\} S_1^* \{R\}$ 和 $\{R\} S_2^* \{Q\}$, 如果 $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{R\} \downarrow \{R\}, \sigma' \rangle$, 那么

$$\langle \{P\} S_1^* \{R\} S_2^* \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{R\} S_2^* \{Q\}, \sigma' \rangle,$$
5. 任给注释程序 $\{P \wedge B\} S_1^* \{Q\}$ 和 $\{P \wedge \neg B\} S_2^* \{Q\}$, 如果 $\sigma \models B$, 则

$$\langle \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P \wedge B\} S_1^* \{Q\}, \sigma \rangle,$$
6. 任给注释程序 $\{P \wedge B\} S_1^* \{Q\}$ 和 $\{P \wedge \neg B\} S_2^* \{Q\}$, 如果 $\sigma \not\models B$, 则

$$\langle \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P \wedge \neg B\} S_2^* \{Q\}, \sigma \rangle,$$
7. 任给 $P \rightarrow I$, 注释程序 $\{I \wedge B\} S^* \{I\}$ 和 $I \wedge \neg B \rightarrow Q$, 如果 $\sigma \models B$, 则

$$\langle \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{I \wedge B\} S^* \{I\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle,$$
8. 任给 $P \rightarrow I$, 注释程序 $\{I \wedge B\} S^* \{I\}$ 和 $I \wedge \neg B \rightarrow Q$, 如果 $\sigma \not\models B$, 则

$$\langle \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} \downarrow \{Q\}, \sigma \rangle.$$

定理 1.2

对任意的 $\langle G, \sigma \rangle \rightarrow \langle G', \sigma' \rangle$,

1. $\text{post}(G') = \text{post}(G)$,
2. 如果 $\text{sk}(G') = \downarrow$, 那么 G 一定形如 $\{P\} \downarrow \{P\}$,
3. $\langle \text{sk}(G), \sigma \rangle \rightarrow \langle \text{sk}(G'), \sigma' \rangle$,
4. 如果 $\sigma \models \text{pre}(G)$, 那么 $\sigma' \models \text{pre}(G')$.

证明.

1. 2. 显然. 3. 4. 对转移操作归纳证明.

i) $P \rightarrow Q$ 且 $G = \{P\} \text{ skip } \{Q\}$, 则 $G' = \{Q\} \downarrow \{Q\}$, $\sigma' = \sigma$. 显然

$$\langle \text{skip}, \sigma \rangle \rightarrow \langle \downarrow, \sigma \rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models Q$.

ii) 有 Q_1 使得 $Q_1[u := t]$ 是自由替换, $P \rightarrow Q_1[u := t]$ 且 $Q_1 \rightarrow Q$, $G = \{P\} u := t \{Q\}$, 则

$$G' = \{Q\} \downarrow \{Q\}, \quad \sigma' = \sigma[u := \sigma(t)],$$

显然 $\langle u := t, \sigma \rangle \rightarrow \langle \downarrow, \sigma[u := \sigma(t)] \rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma \models Q_1[u := t]$, 从而 $\sigma' = \sigma[u := t] \models Q_1$, 即 $\sigma' \models Q$.

iii) $G = \{P\} S_1^* \{R\} S_2^* \{Q\}$, 假设 $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{P'\} S_1'^* \{R\}, \sigma' \rangle$ 并且

$$\langle S_1, \sigma \rangle \rightarrow \langle S_1', \sigma' \rangle, \quad \sigma \models P \rightarrow \sigma' \models P',$$

那么 $G' = \{P'\} S_1'^* \{R\} S_2^* \{Q\}$, 此时 $\langle S_1; S_2, \sigma \rangle \rightarrow \langle S_1'; S_2, \sigma' \rangle$.

iv) $G = \{P\} S_1^* \{R\} S_2^* \{Q\}$, 假设 $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{R\} \downarrow \{R\}, \sigma' \rangle$ 并且

$$\langle S_1, \sigma \rangle \rightarrow \langle \downarrow, \sigma' \rangle, \quad \sigma \models P \rightarrow \sigma' \models R,$$

那么 $G' = \{R\} S_2^* \{Q\}$, 此时 $\langle S_1; S_2, \sigma \rangle \rightarrow \langle S_2, \sigma' \rangle$.

v) $G = \{P\} \text{ if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}$ 且 $\sigma \models B$, 则

$$G' = \{P \wedge B\} S_1^* \{Q\}, \quad \sigma' = \sigma,$$

显然 $\langle \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi}, \sigma \rangle \rightarrow \langle S_1, \sigma \rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma' = \sigma \models P \wedge B$.

vi) $G = \{P\} \text{ if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}$ 且 $\sigma \not\models B$, 则

$$G' = \{P \wedge \neg B\} S_2^* \{Q\}, \quad \sigma' = \sigma,$$

显然 $\langle \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi}, \sigma \rangle \rightarrow \langle S_2, \sigma \rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma' = \sigma \models P \wedge \neg B$.

vii) $P \rightarrow I$, $I \wedge \neg B \rightarrow Q$, $G = \{P\} \{\text{inv} : I\} \text{ while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}$ 且 $\sigma \models B$, 则

$$G' = \{I \wedge B\} S^* \{I\} \{\text{inv} : I\} \text{ while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle \text{while } B \text{ do } S \text{ od}, \sigma \rangle \rightarrow \langle S; \text{while } B \text{ do } S \text{ od}, \sigma \rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models I$, 从而 $\sigma' = \sigma \models I \wedge B$.

viii) $P \rightarrow I$, $I \wedge \neg B \rightarrow Q$, $G = \{P\} \{\text{inv} : I\} \text{ while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}$ 且 $\sigma \not\models B$, 则

$$G' = \{Q\} \downarrow \{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle \text{while } B \text{ do } S \text{ od}, \sigma \rangle \rightarrow \langle \downarrow, \sigma \rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models I$, 从而 $\sigma' = \sigma \models I \wedge \neg B$, 即 $\sigma' \models Q$. □

2 完全正确性注释程序

定义 2.1 (完全正确性注释程序)

对断言 P, Q , 程序 S 和某字符串 S^* , 递归定义

$$\{P\} S^* \{Q\} \vdash_{\text{TO}} [P] S [Q],$$

称 $\{P\} S^* \{Q\}$ 是 $[P] S [Q]$ 的**完全正确性注释程序**, 前后断言与骨架程序的记法 pre, post, sk 同上节:

1. 当 $P \rightarrow Q$ 时, $\{P\} \text{skip} \{Q\} \vdash_{\text{TO}} [P] \text{skip} [Q]$,
2. 当 $P \rightarrow Q[u := t]$ 时, $\{P\} u := t \{Q\} \vdash_{\text{TO}} \{P\} u := t \{Q\}$,
3. 如果 $\{P\} S_1^* \{R\} \vdash_{\text{TO}} [P] S_1 [R]$ 且 $\{R\} S_2^* \{Q\} \vdash_{\text{TO}} [R] S_2 [Q]$, 则

$$\{P\} S_1^* \{R\} S_2^* \{Q\} \vdash_{\text{TO}} [P] S_1; S_2 [Q],$$

4. 如果 $\{P \wedge B\} S_1^* \{Q\} \vdash_{\text{TO}} [P \wedge B] S_1 [Q]$ 且 $\{P \wedge \neg B\} S_2^* \{Q\} \vdash_{\text{TO}} [P \wedge \neg B] S_2 [Q]$, 则

$$\begin{aligned} & \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\} \\ & \vdash_{\text{TO}} [P] \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } [Q], \end{aligned}$$

5. 如果 $P \rightarrow I$, $\{I \wedge B\} S^* \{I\} \vdash_{\text{TO}} [I \wedge B] S [I]$, t 是整型项, z 是整型变量, z 不出现在 t, B, I, S 中, $\{I \wedge B \wedge t = z\} S^{**} \{t < z\} \vdash_{\text{TO}} [I \wedge B \wedge t = z] S [t < z]$, $I \rightarrow t \geq 0$ 且 $I \wedge \neg B \rightarrow Q$, 则

$$\begin{aligned} & \{P\} \{\text{inv} : I\} \{\text{bd} : t\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\} \\ & \vdash_{\text{TO}} [P] \text{while } B \text{ do } S \text{ od } [Q]. \end{aligned}$$

定理 2.1 (正规形定理)

对任意的断言 P, Q 和程序 S , $\vdash_{\text{TW}} [P] S [Q]$ 当且仅当存在 $\{P\} S^* \{Q\} \vdash_{\text{TO}} [P] S [Q]$.

完全正确性注释程序上的操作几乎与定义 1.1 相同 (只是多了 **bd** 标记), 定理 1.2 的四条性质对完全正确性注释程序也依旧成立.